



Practice Problem Set -03 (Taylor's expansions & Maclaurin's expansions)

1. Using Maclaurin's series, expand the following functions:

i) $\sin^{-1} x$

ii) $\tan^{-1} x$

iii) $\log \sec x$

iv) $e^x \sin x$

2. Expand $\sin x$ in powers of $\left(x - \frac{\pi}{2}\right)$

Hence find the value of $\sin 91^\circ$ correct to 4 decimal places.

3. Prove that $\log \sin x = \log \sin a + (x - a) \cot a - \frac{1}{2} (x - a)^2 \operatorname{cosec}^2 a + \dots$

4. Find the Taylor's series expansion for (i) $\frac{1}{x}$ about 1. ii) $\log \cos x$ about the point $\frac{\pi}{3}$.

5. Compute to four decimal places, the value of $\cos 32^\circ$, by the use of Taylor's series.

6. Calculate approximately i) $\log_{10} 404$, given $\log 4 = 0.6021$. ii) $(1.04)^{3.01}$